

Switching Concepts — Guided Notes ANSWER KEY

Unit 2.1.1 | CompTIA Network+ N10-009 Obj. 2.1 | N10-008 2.1

For instructor use / distribute after students complete the notes.

Question 1

A switch learns MAC addresses by reading the _____ on every incoming frame and recording which _____ it arrived on. Entries age out after typically _____ seconds. This table is also called the _____ table.

Answer: Source MAC address; port; 300 seconds; CAM (Content Addressable Memory).

Question 2

Match each switching behavior to its condition:

Forwarding → _____

Filtering → _____

Flooding → _____

Answer: Forwarding → destination MAC is in the table on a different port, send out that port only; Filtering → destination MAC is on the same port the frame arrived on, drop it; Flooding → destination MAC is unknown or it's a broadcast, send out all ports except the source port.

Question 3

Each switch port is its own _____ domain, eliminating collisions. By default, the entire switch is one _____ domain — a broadcast reaches every port. _____ and _____ are the two tools used to break broadcast domains.

Answer: Collision; broadcast; Routers; VLANs.

Question 4

Full duplex allows a device to _____ and _____ simultaneously, effectively doubling available bandwidth. A duplex mismatch occurs when one side negotiates _____ and the other negotiates _____. The symptom is _____.

Answer: Transmit; receive; full duplex; half duplex; high latency, excessive errors, late collisions — connection works but performs very poorly.

Question 5

Without STP, redundant switch links create _____. A broadcast frame enters the loop and circulates _____, multiplying at every junction. This is called a _____ and will take down the network in seconds.

Answer: Layer 2 loops; forever; broadcast storm.

Question 6

In STP, the switch with the lowest _____ becomes the root bridge. The Bridge ID is composed of _____ and _____. In production, you should set _____ manually to control which switch wins

the election.

Answer: Bridge ID; priority value; MAC address; priority.

Question 7

A port in STP _____ state receives BPDUs but does not forward frames. The transition from blocking to forwarding in classic STP takes approximately _____ seconds. RSTP reduces this to _____ and consolidates port states to: _____, _____, and _____.

Answer: Blocking; 30-50; seconds (sub-second to a few seconds); Discarding, Learning, Forwarding.

Question 8 — Real World Application

REAL WORLD: A user plugs a cheap unmanaged switch into a wall jack to add more ports at their desk. Within seconds, the entire floor loses network connectivity. Explain what happened and what configuration would have prevented it.

Answer: The unmanaged switch created a Layer 2 loop — its ports connected back to the network in multiple places with no STP awareness, causing a broadcast storm. Prevention: enable PortFast on all access ports (so end devices come up immediately) combined with BPDU Guard (which shuts down the port the instant it receives a BPDU from a connected switch). BPDU Guard would have shut down the port when the unmanaged switch sent a BPDU, containing the damage immediately.